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CMSC 207 CRN 32950

CMSC 207: Chapter 1 Homework

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1. The functions F and G from $\mathbb{R} \to \mathbb{R}$ are defined by the following formulas:

$$F(x) = (x+1)(x-3) \qquad\qquad G(x) = (x-2)^2 - 7$$

Prove that $F \neq G$.

$F \neq G$ means $\exists x \in \mathbb{R}$ such that $F(x) \neq G(x)$. Choose $x = 1 \in \mathbb{R}$. Then:

$$\begin{aligned} F(x) &= F(1) = (1+1)(1-3) = 2(-2) &&= -4. \\ G(x) &= G(1) = (1-2)^2 - 7 = (-1)^2 - 7 = 1 - 7 &&= -6. \end{aligned}$$

Since $F(1) = -4$ and $G(1) = -6$ and $-4 \neq -6$, we have $F(1) \neq G(1)$. Thus, $F \neq G$. ∎

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2. The relation R from $\mathbb{R} \to \mathbb{R}$ is defined as follows:

$$\text{for all } (x, y) \in \mathbb{R} \times \mathbb{R}, (x, y) \in R \text{ iff } x = y^2 + 1.$$

Prove that $(17, -4) \in R$.

By the definition of R,

$$(x, y) \in R \iff x = y^2 + 1.$$

Substituting $(x, y) = (17, -4) \in \mathbb{R} \times \mathbb{R}$ we have:

$$\begin{aligned}
(17, -4) \in R \iff 17 &= (-4)^2 + 1 \\
&= 16 + 1 && \text{by arithmetic of rhs} \\
&= 17. && \text{by arithmetic of rhs}
\end{aligned}$$

Since the $17 = 17$ is a true statement, $(17, -4) \in R$ must also be true. ∎